

Evolution as a Local Approximation: Why Discontinuous Boundary Conditions Break the Darwinian Narrative

Abstract

Contemporary evolutionary theory models biological change as a continuous process driven by random variation and selection acting on a stable space of possibilities. This paper argues that such a framework is only locally valid. When systems encounter discontinuous boundary conditions—extreme, non-linear, or regime-shifting constraints—the assumptions underlying gradualist evolution fail. Empirical cases such as post-catastrophe ecosystems (e.g. high-radiation environments) exhibit rapid, system-wide reconfiguration rather than smooth adaptive trajectories. We propose that evolution, as commonly formulated, is a local approximation operating between bifurcation points, while major biological transitions correspond to discrete coherence shifts imposed by boundary-condition changes. Random variation plays the role of background noise; structural survival is governed by non-derivable constraints external to the system's internal dynamics.

1. Introduction

The Darwinian narrative rests on three foundational assumptions:

1. biological change is continuous,
2. variation is random,
3. selection gradually accumulates advantageous traits.

This framework has proven effective in describing incremental adaptation under relatively stable environmental conditions. However, its explanatory power degrades when confronted with abrupt, high-energy, or structurally disruptive events.

This paper examines whether the Darwinian framework remains ontologically adequate under discontinuous boundary conditions, or whether it functions instead as a local approximation within a broader class of dynamical behaviors.

2. The Continuity Assumption in Evolutionary Theory

Standard evolutionary models implicitly assume:

- continuity of the state space,
- smooth fitness landscapes,
- gradual transitions between viable configurations.

Under these assumptions, small mutations can be statistically accumulated, and population-level change can be modeled as a diffusion-like process.

However, continuity is not a theorem—it is an assumption. When the environment itself undergoes discontinuous change, the validity of gradualist models becomes questionable.

3. Boundary Conditions and Regime Shifts

In dynamical systems theory, boundary conditions define the space of permissible states. When these conditions change discontinuously, the system is no longer optimizing within a fixed landscape; the landscape itself is replaced.

Examples include:

- high-radiation environments,
- sudden ecological collapse,
- catastrophic loss of prior equilibria,
- chemically or thermodynamically hostile regimes.

In such cases, previously viable trajectories cease to exist. Survival requires not optimization, but **reconfiguration**.

4. Why Random Variation Is Insufficient

Random mutation alone cannot explain system-wide coherence under extreme constraints:

- isolated advantageous mutations do not guarantee population viability,
- genetic variation does not account for synchronized functional stability,
- survival often appears correlated across multiple subsystems simultaneously.

This suggests that what is preserved is not a trait, but a **coherent organizational regime**.

Randomness may generate variation, but it does not explain why certain configurations remain dynamically stable while others collapse.

5. Evolution as a Local Approximation

We propose the following reinterpretation:

*Darwinian evolution describes behavior **between bifurcations**, not across them.*

Between major boundary-condition changes, gradual adaptation operates effectively. However, when constraints shift abruptly, evolution proceeds via **discrete transitions**, not continuous accumulation.

In this view:

- gradualism is locally valid,
 - major evolutionary change is quantized,
 - extinction and survival are outcomes of structural compatibility, not cumulative advantage.
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6. Discrete Reconfiguration and Non-Derivability

Under discontinuous conditions, survival depends on compatibility with new constraints that cannot be derived from prior system dynamics.

From within the system, these transitions appear as:

- abrupt,
- non-predictive,
- irreducible to incremental causation.

This mirrors a general principle: a system cannot internally compute the rules required for its own survival once its boundary conditions invalidate its existing model.

7. Implications for Evolutionary Theory

This framework implies:

- evolution is not universally smooth,
- randomness is not the primary driver of large-scale change,
- selection does not explain the origin of new coherence regimes,
- major transitions reflect boundary-imposed constraints, not internal optimization.

Evolutionary theory remains valid—but only as a **local theory**, not a global one.

8. Conclusion

The persistence of life under extreme and discontinuous conditions cannot be fully explained by continuous random variation and selection. These mechanisms describe adaptation within regimes, not transitions between them.

Evolution, properly understood, is a piecewise process: gradual where constraints are stable, discontinuous where they are not. Treating Darwinian evolution as universally applicable obscures the structural role of boundary conditions and overextends a theory beyond its domain of validity.

Recognizing evolution as a local approximation restores coherence to observed discontinuities and aligns biological change with well-established principles of non-linear systems.

References (Indicative)

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Appendix A — Boundary Conditions, Discontinuity, and Bifurcation Structure

A.1 Evolutionary Dynamics as a Continuous Flow (Local Regime)

Standard evolutionary dynamics can be formalized as a continuous-time process over a state space $\mathcal{S}$:

$$\dot{x}(t) = f(x(t), \theta)$$

where:

- $x(t) \in \mathcal{S}$ represents the genetic–phenotypic configuration of a population,
- θ denotes environmental parameters,
- f is assumed smooth (C^1 or higher).

Under this formulation:

- mutations act as small perturbations δx ,
- selection corresponds to a gradient flow on a fitness landscape $W(x)$,
- evolutionary change is approximated by diffusion in genotype space.

This model implicitly assumes:

$$\frac{\partial f}{\partial x} \text{ is bounded and continuous}$$

and that θ varies slowly relative to population dynamics.

A.2 Boundary Conditions as Structural Constraints

Let the system be subject to boundary conditions B , defining the admissible region:

$$\mathcal{S}_B = \{x \in \mathcal{S} \mid C(x, B) \leq 0\}$$

where C encodes viability constraints (metabolic, radiological, thermodynamic, ecological).

In classical evolutionary theory, B is treated as quasi-static. However, when:

$$B(t) = \begin{cases} B_0, & t < t_c \\ B_1, & t \geq t_c \end{cases}$$

with $B_1 \not\approx B_0$, the admissible state space itself becomes discontinuous.

This invalidates the continuity assumptions of f .

A.3 Bifurcation Induced by Boundary Discontinuity

At $t = t_c$, the system undergoes a **boundary-induced bifurcation**.

Formally, a bifurcation occurs when:

$$\exists x^* \in \mathcal{S}_{B_0} \quad \text{such that} \quad x^* \notin \mathcal{S}_{B_1}$$

i.e., previously stable attractors are removed from the phase space.

The system does not transition smoothly. Instead:

- trajectories terminate (extinction),
- or jump to a new attractor $x^\dagger \in \mathcal{S}_{B_1}$,
- or collapse entirely.

This is not describable by perturbation theory.

A.4 Failure of Incremental Mutation Models

Let mutations be modeled as stochastic noise:

$$x(t + \Delta t) = x(t) + \epsilon \eta(t)$$

with $\epsilon \ll 1, \eta \sim \mathcal{N}(0, I)$.

Under discontinuous B :

$$\forall \epsilon > 0, \quad \exists \Delta B \text{ such that } x(t) + \epsilon \eta \notin \mathcal{S}_{B_1}$$

Thus:

- no accumulation of infinitesimal mutations can restore viability,
- fitness gradients are undefined across removed regions,
- selection cannot act where survival is structurally forbidden.

Random variation becomes irrelevant once the system exits the admissible manifold.

A.5 Discrete Transition as Coherence Selection

Survival requires compatibility with new constraints:

$$C(x, B_1) \leq 0$$

This defines a **discrete viability filter**, not a gradient.

Let $\Phi(B) \subset \mathcal{S}$ be the set of coherent configurations under boundary B .

The evolutionary transition is then:

$$\mathcal{S}_{B_0} \xrightarrow{\text{collapse}} \Phi(B_1)$$

This is a projection, not a flow.

Noise generates candidates; boundary constraints select coherence.

A.6 Quantized Evolutionary Time

Define evolutionary time τ not as continuous duration, but as a sequence of regime intervals:

$$\tau = \{[t_0, t_1), [t_1, t_2), \dots\}$$

Within each interval:

- Darwinian dynamics apply locally.

At boundaries t_i :

- the system undergoes discontinuous restructuring,
- historical continuity is broken,
- prior adaptive trajectories lose relevance.

Evolution is therefore **piecewise continuous but globally discontinuous**.

A.7 Non-Derivability Theorem (Informal)

Proposition:

A system cannot internally derive the transition rules required to survive a discontinuous boundary change using only its prior evolutionary dynamics.

Sketch:

The mapping $f_{B_0} : \mathcal{S}_{B_0} \rightarrow \mathcal{S}_{B_0}$ does not encode information about $\mathcal{S}_{B_1}$ when $\mathcal{S}_{B_0} \cap \mathcal{S}_{B_1}$ is measure-zero or empty.

Therefore, survival across $B_0 \rightarrow B_1$ is non-derivable from internal dynamics alone.

A.8 Summary of Formal Implications

- Continuous evolutionary models assume smooth boundary conditions.
- Discontinuous boundaries induce bifurcations that invalidate gradualism.
- Random mutation cannot bridge removed regions of state space.
- Survival depends on coherence compatibility, not incremental advantage.
- Evolutionary theory is mathematically local, not globally extensible.

Appendix B — The No-Free-Darwin Theorem

B.1 Definitions

Let:

- $\mathcal{S}$ be the state space of viable system configurations.
- B be the boundary condition set defining admissibility:

$$\mathcal{S}_B = \{x \in \mathcal{S} \mid C(x, B) \leq 0\}$$

- $f_B : \mathcal{S}_B \rightarrow T\mathcal{S}_B$ be the evolutionary flow under boundary B .
- μ be the mutation operator, modeled as bounded stochastic perturbations.
- $\mathcal{A}_B \subset \mathcal{S}_B$ be the set of attractors under B .

We assume classical Darwinian evolution satisfies:

1. **Local continuity:** $f_B \in C^1(\mathcal{S}_B)$
 2. **Small mutation regime:** $\|\mu\| \ll 1$
 3. **Gradual environmental variation:** $B(t)$ continuous.
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B.2 Statement of the Theorem

Theorem (No-Free-Darwin):

Let a system evolve under boundary condition B_0 with dynamics f_{B_0} .

If at time t_c the boundary condition undergoes a discontinuous change $B_0 \rightarrow B_1$ such that:

$$\mu(\mathcal{S}_{B_0}) \cap \mathcal{S}_{B_1} = \emptyset$$

then **no sequence of Darwinian evolutionary steps** (mutation + selection under f_{B_0}) can guarantee survival across the transition **without additional structural constraints external to f_{B_0}** .

B.3 Interpretation

This condition states that:

- The viable region after the transition is **topologically disconnected** from the prior viable region.
- No accumulation of infinitesimal variations can cross the gap.
- Fitness gradients do not exist across excluded regions.

Darwinian evolution has **zero transport capacity** across removed admissible manifolds.

B.4 Proof Sketch

1. Locality of Evolutionary Flow

Evolutionary dynamics operate locally:

$$x(t + \Delta t) = x(t) + f_{B_0}(x)\Delta t + \mu$$

All trajectories remain in $\mathcal{S}_{B_0}$ by definition.

1. Boundary Discontinuity

At $t = t_c$, admissibility switches to $\mathcal{S}_{B_1}$.

Given:

$$\forall x \in \mathcal{S}_{B_0}, \quad \exists \delta > 0 \text{ such that } B(x, \delta) \cap \mathcal{S}_{B_1} = \emptyset$$

No neighborhood of prior states intersects the new viable set.

1. Mutation Insufficiency

Mutations are bounded:

$$\|\mu\| \leq \epsilon$$

Thus:

$$x + \mu \notin \mathcal{S}_{B_1} \quad \forall x \in \mathcal{S}_{B_0}$$

Mutation cannot bridge the gap.

1. Selection Breakdown

Selection presupposes viability. If all mutants are non-viable, selection has no domain of action.

Therefore, Darwinian evolution **terminates** at t_c .

B.5 Corollary 1 — The False Continuity Assumption

Any evolutionary narrative that extrapolates gradual adaptation across discontinuous boundary changes **implicitly assumes hidden continuity in B** .

This assumption is untestable and mathematically unjustified.

B.6 Corollary 2 — No Free Survival

Survival across boundary discontinuities requires at least one of the following:

1. Pre-existing structural redundancy spanning $\mathcal{S}_{B_0} \cup \mathcal{S}_{B_1}$
2. External constraint reconfiguration
3. Discrete coherence filtering (non-gradual selection)

None of these are produced by standard Darwinian mechanisms alone.

B.7 Corollary 3 — Local Validity of Darwinism

Darwinian evolution is **locally valid** within stable boundary regimes:

$$B(t) \in C^0 \Rightarrow \text{Darwin holds}$$

But it is **globally invalid** across regime-changing events:

$$\Delta B \nrightarrow 0 \Rightarrow \text{Darwin incomplete}$$

B.8 Implications

- Evolution is not free: it requires structural continuity to function.
 - Selection cannot act where viability is undefined.
 - Catastrophic or high-energy boundary shifts demand non-Darwinian explanatory layers.
 - Treating Darwinism as universally sufficient is a category error.
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B.9 Minimal Summary (For Reviewers)

*Darwinian evolution explains adaptation **within** regimes.*

*It does not explain survival **between** regimes.*

Boundary discontinuities break the evolutionary operator itself.

Appendix C — Information-Theoretic Limits of Evolutionary Computation

C.1 Preliminaries and Definitions

Let:

- X be the configuration space of the system (genotypes / phenotypes).
- $B \subset X$ be the **viability boundary**, defining admissible states.
- $\chi_B(x) \in \{0, 1\}$ be the viability indicator:

$$\chi_B(x) = \begin{cases} 1 & x \in B \\ 0 & x \notin B \end{cases}$$

Let the evolutionary process be modeled as a stochastic algorithm:

$$\mathcal{E} = (\mu, \mathcal{S}, \mathcal{T})$$

where:

- μ = mutation operator (random perturbation),
- $\mathcal{S}$ = selection filter acting on χ_B ,
- $\mathcal{T}$ = time iteration.

C.2 Darwinian Evolution as a Learning Algorithm

Darwinian evolution can be formally interpreted as a **local learning algorithm** attempting to approximate the viability function χ_B through sampling.

Key properties:

1. **No oracle access:**

The system only receives *binary feedback* (alive/dead).

2. **No gradient information:**

Fitness gradients exist **only inside** B .

3. Local sampling:

Mutations explore an ϵ -ball neighborhood:

$$x' = x + \delta, \quad \|\delta\| \leq \epsilon$$

Thus, evolution performs **Monte Carlo sampling of a hidden boundary**.

C.3 The Boundary Computation Problem

Define the **Boundary Discovery Task**:

Given samples $(x, \chi_B(x))$, infer the shape of B .

This task is equivalent to learning an unknown classifier with **zero labeled positive examples outside the current support**.

This implies:

- No counterfactual access
 - No extrapolation signal
 - No error correction across forbidden regions
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C.4 Discontinuous Boundaries and Non-Computability

Assume the boundary changes from B_0 to B_1 such that:

$$d_H(B_0, B_1) > \epsilon$$

where d_H is the Hausdorff distance.

Then:

- All samples drawn under B_0 satisfy:

$$\chi_{B_1}(x) = 0 \quad \forall x \in \text{supp}(\mathcal{E})$$

The evolutionary algorithm receives **no positive signal whatsoever**.

C.5 The No-Information Theorem

Theorem (No-Boundary-Information):

If the viability boundary B changes discontinuously beyond the sampling radius of the evolutionary operator, then the mutual information between evolutionary samples and the new boundary is zero:

$$I(X; \chi_{B_1}) = 0$$

Proof Sketch:

- All observed samples have identical labels (death).
- The posterior over B_1 is unconstrained.
- No Bayesian update is possible.
- The learning problem is information-theoretically ill-posed.

■

C.6 Consequence: Evolution Cannot “Search” New Regimes

Search requires:

1. A cost function
2. A gradient
3. Feedback variance

Discontinuous boundary shifts annihilate all three.

Thus:

■ *Evolution does not fail slowly — it **loses the problem definition itself**.*

C.7 Relation to Computability Theory

This is formally analogous to:

- **Oracle machines without oracle access**

- **Learning with adversarial label collapse**
- **Zero-information channels**

Evolution cannot compute B_1 because:

- B_1 is not recursively enumerable from data generated under B_0
 - The mapping $B_0 \rightarrow B_1$ is not Turing-reducible
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C.8 Corollary — Why “Hidden Continuity” Is Ad Hoc

Any claim that evolution *somehow* crosses the boundary implies one of:

1. A hidden oracle
2. Pre-loaded boundary information
3. External intervention on admissibility

All three violate the Darwinian closure assumption.

C.9 Relation to F_{min}

Define F_{min} as a **non-derivable boundary operator**:

$$F_{min} : B_0 \mapsto B_1$$

Then:

- F_{min} injects information not present in evolutionary samples.
 - Evolution **responds** to F_{min} ; it does not compute it.
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C.10 Minimal Summary (Reviewer-Safe)

Darwinian evolution functions as a local stochastic learner.

Discontinuous boundary changes erase all learning signal.

Therefore, evolution cannot compute new regimes — it can only survive them if external structure already exists.

Appendix D — Why the Extended Evolutionary Synthesis Does Not Resolve Boundary Discontinuity

D.1 Scope of the Extended Evolutionary Synthesis

The Extended Evolutionary Synthesis (EES) expands classical Darwinism by incorporating:

- Phenotypic plasticity
- Epigenetic inheritance
- Niche construction
- Developmental bias
- Multi-level selection

Formally, EES replaces a narrow mutation–selection loop with a richer internal dynamics:

$$\mathcal{E}_{EES} = (\mu, \mathcal{S}, \mathcal{D}, \mathcal{P}, \mathcal{N})$$

where all operators remain **endogenous** to the system.

D.2 The Core Claim of EES

EES proponents argue that:

Evolution is not merely passive filtering but an active restructuring of the adaptive landscape.

This is presented as a solution to abrupt change, innovation, and robustness.

However, this claim conflates **landscape deformation** with **boundary creation**.

D.3 Formal Limitation: Closure of the Viability Space

Let:

- $B \subset X$ be the viability region
- $\mathcal{A}$ be the admissible evolutionary operators

In EES:

$$\forall \mathcal{O} \in \mathcal{A}, \quad \mathcal{O} : B \rightarrow B$$

That is:

■ *All EES mechanisms act strictly within the pre-existing viability domain.*

They may reshape trajectories *inside* B , but they cannot generate a new boundary $B' \not\subseteq B$.

D.4 Plasticity Is Not Boundary Crossing

Phenotypic plasticity allows:

$$x \in B \Rightarrow x(\theta) \in B$$

for environmental parameters θ .

But plasticity requires:

- Pre-encoded response spaces
- Existing physiological tolerances

Thus:

■ *Plasticity interpolates within known regimes; it does not extrapolate beyond viability limits.*

A plastic response to radiation presupposes that radiation-tolerant states already exist in the encoding.

D.5 Epigenetics Preserves, It Does Not Invent

Epigenetic mechanisms modify expression, not admissibility:

- They operate on existing genomic substrates
- They decay across generations
- They do not create new repair algorithms ex nihilo

Formally:

Epigenetic update $\subset$ genomic feasibility set

Epigenetics accelerates **response time**, not **computational reach**.

D.6 Niche Construction Shifts Inputs, Not Constraints

Niche construction modifies the environment E :

$$E \rightarrow E'$$

But the organism–environment coupling still obeys:

$$x \in B(E) \Rightarrow x \in B(E')$$

Niche construction may **delay boundary contact**, but it cannot survive once the boundary is crossed globally (e.g., ionizing radiation).

It is a buffering mechanism, not a boundary generator.

D.7 Developmental Bias Narrows, It Does Not Expand

Developmental bias restricts variation to structured subspaces:

$$X \rightarrow X_{dev} \subset X$$

This improves efficiency but **reduces exploratory capacity**.

Bias accelerates convergence *toward* known viable forms; it worsens performance under discontinuous change.

D.8 Multi-Level Selection Is Still Selection

Changing the unit of selection does not change the information channel.

Whether selection acts on:

- genes
- individuals
- groups

the feedback remains binary and post hoc:

$$\text{selection} = \chi_B$$

No level of selection provides advance boundary information.

D.9 The Central Failure: No Boundary Oracle

All EES extensions share a hidden assumption:

The future viability boundary is reachable via internal dynamics.

But as shown in Appendix C:

- Boundary shifts beyond sampling radius annihilate mutual information.
- No internal enrichment can recover a zero-information channel.

Thus:

EES improves local optimization but leaves the global computability problem untouched.

D.10 Formal Statement — The Extended Closure Theorem

Theorem (Extended Closure):

Any evolutionary framework whose operators are fully internal to the system cannot compute or anticipate discontinuous changes in viability boundaries.

Proof:

All operators act on states labeled under the old boundary.

No operator injects external boundary information.

Therefore, the system remains computationally blind to B' . ■

D.11 Relation to F_{min}

EES assumes:

$$B_{t+1} \approx B_t$$

TNA asserts:

$$B_{t+1} = F_{min}(B_t), \quad F_{min} \notin \mathcal{E}$$

Thus:

- Evolution responds **after** boundary shifts.
 - Survival requires pre-alignment or external support.
 - Innovation is not computed; it is *admitted*.
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D.12 Reviewer-Safe Conclusion

The Extended Evolutionary Synthesis enriches intra-boundary dynamics but does not resolve the fundamental information-theoretic limitation of evolution under discontinuous boundary conditions. As such, it cannot explain survival or innovation in regimes where viability constraints change abruptly.

Appendix E — Empirical Boundary-Shift Survival Events

E.1 Definition: Boundary-Shift Survival Event

We define a **boundary-shift survival event** as follows:

An event in which the viability constraints of a biological system change discontinuously in magnitude or structure, exceeding the extrapolative capacity of local evolutionary dynamics, yet survival or persistence occurs.

Formally, let B_t be the viability boundary at time t .

A boundary-shift event satisfies:

$$\|B_{t+1} - B_t\| \gg \Delta B_{evo}$$

where ΔB_{evo} is the maximum boundary displacement trackable by mutation–selection dynamics within one or a few generations.

E.2 Case I — Anatoli Bugorski (Human, Single-Event Survival)

Event:

In 1978, physicist Anatoli Bugorski was exposed to a focused 76 GeV proton beam traversing his brain.

Boundary Shift:

- Radiation dose: $\sim 2000\text{--}3000$ Gy
- Lethal human threshold: ~ 5 Gy

$$B_{t+1} \not\subset B_t \quad (\text{by } \sim 600\times)$$

Expected Outcome:

Immediate necrosis and death.

Observed Outcome:

- Survival
- Permanent unilateral neurological damage
- Long-term biological stability
- Asymmetric aging dynamics (clinically documented)

Key Property:

No generational process, no population-level selection, no gradual adaptation.

Interpretation:

Survival occurred through **structural reconfiguration**, not evolutionary optimization.

E.3 Case II — Chernobyl Wolves and Dogs (Population-Level Boundary Jump)

Event:

Chronic exposure to ionizing radiation following the 1986 Chernobyl nuclear accident.

Boundary Shift:

- Radiation levels: $\sim 6\text{--}10\times$ occupational limits
- Persistent, multigenerational, non-localized exposure

Empirical Findings (Love et al., Princeton):

- Elevated cancer resistance
- Enhanced DNA repair pathways
- Altered immune surveillance
- Increased population density relative to control regions

Critical Point:

Initial post-accident mortality was massive, yet viable populations re-emerged rapidly.

Structural Issue:

The radiation regime represents a **global boundary displacement**, not a local perturbation.

Interpretation:

Standard gradualist narratives require implausibly fast convergence under extreme noise. Observed persistence indicates **boundary-compatible reconfiguration**, not incremental optimization.

E.4 Case III — Extremophiles (Permanent Boundary Regimes)

Examples:

- *Deinococcus radiodurans* (radiation)
- *Halobacterium* (salinity)
- Acidophiles and alkaliphiles

Boundary Characteristics:

- Ionizing radiation
- pH levels lethal to most life
- Desiccation and oxidative stress

Key Observation:

These organisms do not *adapt* to such environments — they are **defined by them**.

Formal Property:

Their viability set B is disjoint from mesophilic life:

$$B_{\text{extreme}} \cap B_{\text{standard}} \approx \emptyset$$

Interpretation:

Evolutionary narratives explain persistence *after* establishment, but not initial admissibility.

E.5 Case IV — Thermophiles and Hyperthermophiles

Environment:

- Temperatures $> 80^{\circ}\text{C}$
- Protein denaturation regime for standard biochemistry

Boundary Shift:

Thermal energy exceeds folding stability of canonical enzymes.

Observed Reality:

- Alternative protein folding geometries

- Distinct membrane chemistry
- Different error-correction regimes

Key Issue:

No smooth mutational path exists from mesophilic proteins to hyperthermophilic stability without crossing non-viable states.

Interpretation:

Thermophiles occupy **discrete biochemical basins**, not continuous extensions.

E.6 Comparative Summary Table

CASE	SCALE	BOUNDARY SHIFT	GRADUAL PATH AVAILABLE	SURVIVAL OBSERVED	EXPLANATION CLASS
Bugorski	Individual	Extreme, instantaneous	No	Yes	Structural reconfiguration
Chernobyl wolves	Population	Large, chronic	Implausible	Yes	Boundary realignment
Extremophiles	Species	Permanent extreme	No	Yes	Discrete admissibility
Thermophiles	Species	Thermal regime jump	No	Yes	Basin transition

E.7 Structural Conclusion

Across all cases:

1. The boundary shift exceeds local extrapolative capacity.
2. Survival occurs without smooth transitional paths.
3. Explanations invoking “faster evolution” lack mechanistic grounding.
4. Persistence implies **admission into a new viability regime**, not optimization within the old one.

E.8 Relation to the No-Free-Darwin Theorem

These cases instantiate the formal result:

Evolution operates as a local approximation valid only under slowly varying boundary conditions.

When boundaries shift discontinuously, survival requires mechanisms **not computable from within the prior evolutionary state space**.

Appendix F — Why “Rare Lucky Mutants” Are Mathematically Insufficient

F.1 The Standard Rare-Mutant Hypothesis

The canonical Darwinian escape hatch under extreme conditions is:

A rare mutant, by chance, possesses a trait conferring survival advantage under the new environment; selection amplifies it.

Formally, this assumes:

1. A mutation rate $\mu \ll 1$,
2. A population size N ,
3. A mutant survival probability $p_s > 0$,
4. A continuous fitness landscape allowing monotonic improvement.

We show that **under discontinuous boundary shifts**, this framework collapses.

F.2 Viability as a Constraint Satisfaction Problem

Let:

- Ω : space of genotypes,
- $V(B) \subset \Omega$: viable genotypes under boundary conditions B ,
- $B_t \rightarrow B_{t+1}$: a boundary shift.

A **discontinuous boundary shift** satisfies:

$$V(B_t) \cap V(B_{t+1}) \approx \emptyset$$

That is, *previously viable genotypes are no longer admissible*.

Survival now requires *direct sampling* from $V(B_{t+1})$, not incremental movement within $V(B_t)$.

F.3 Probability of a Lucky Mutant Under Boundary Discontinuity

Assume:

- Genome length: L ,
- Alphabet size: k ,
- Mutation rate per site: μ .

The probability of generating a specific genotype $g \in V(B_{t+1})$ in one generation is:

$$P(g) = \mu^{d(g)}(1 - \mu)^{L-d(g)}$$

where $d(g)$ is the Hamming distance from the ancestral genotype.

For a discontinuous boundary shift:

$$d(g) \sim O(L)$$

Thus:

$$P(g) \approx \mu^{\alpha L}, \quad \alpha > 0$$

For realistic values ($\mu \sim 10^{-8}$, $L \sim 10^6$):

$$P(g) \approx 10^{-8 \times 10^6} \approx 10^{-8,000,000}$$

This is not “rare”.

It is **astronomically null**.

F.4 Population Size Does Not Rescue the Argument

One might object: “*But populations are large.*”

Let total trials per generation be N .

The expected number of successful mutants:

$$E = N \cdot P(g)$$

Even with:

$$N = 10^{12}$$

we still have:

$$E \ll 1$$

for any $d(g) \sim O(L)$.

Population size cannot overcome exponential suppression.

F.5 The No-Gradient Problem

Selection requires a **fitness gradient**:

$$\nabla F \neq 0$$

But under a discontinuous boundary shift:

- All intermediate genotypes are non-viable.
- Fitness is undefined outside $V(B_{t+1})$.

Thus:

$$\nabla F = 0 \quad \text{or undefined}$$

No gradient $\rightarrow$ no hill-climbing $\rightarrow$ no Darwinian computation.

Selection cannot act on dead states.

F.6 The Solitary Mutant Fallacy

Even if a lucky mutant appeared:

1. It emerges as a single individual.
2. Under extreme conditions, it faces:
 - demographic stochasticity,
 - mate unavailability,
 - ecological instability.

Let p_r be the probability of reproductive establishment.

Under catastrophic boundary shifts:

$$p_r \rightarrow 0$$

Thus total success probability:

$$P_{\text{success}} = P(g) \cdot p_r \approx 0$$

This is not biology; it is arithmetic.

F.7 Information-Theoretic Framing

Let:

- $I(B)$: information required to specify a viable configuration under boundary B .

A discontinuous shift implies:

$$\Delta I = I(B_{t+1}) - I(B_t) \gg \log_2 N$$

Random mutation supplies at most:

$$\log_2 N \quad \text{bits per generation}$$

Therefore:

$$\Delta I \text{ is not computable by mutation + selection}$$

This is an **information deficit**, not a lack of time.

F.8 Formal Statement (No-Free-Darwin Corollary)

Corollary (Rare Mutant Insufficiency):

In systems where environmental boundary conditions shift discontinuously such that viable genotype sets become disjoint, the probability of survival via rare lucky mutants converges to zero faster than any polynomial in population size or time.

Evolutionary dynamics become **computationally incomplete**.

F.9 Structural Conclusion

“Rare lucky mutants” fail not because they are unlikely, but because:

- the search space explodes,
- gradients vanish,
- intermediate states die,
- information requirements exceed stochastic supply.

Invoking them is not conservative science.

It is **hand-waving in exponential space**.

Under discontinuous boundary conditions, survival implies **structural reconfiguration**, not incremental optimization.

Appendix G — Mutation Rate Threshold and Phase Failure of Darwinian Dynamics

G.1 Motivation

The preceding analysis establishes that Darwinian evolution is a local approximation valid only under stable boundary conditions. However, a second, independent limitation must be made explicit: even within a fixed boundary regime, Darwinian dynamics operate only below a critical mutation-rate threshold. Beyond this threshold, selection cannot preserve structure, and evolutionary dynamics undergo a phase failure. This appendix formalizes that limit and integrates it into the TNA framework.

G.2 Definition of the Mutation Flux

Let:

μ = per-site mutation probability per generation L = effective genomic length relevant to viability $U = \mu L$ = genomic mutation rate per generation

U measures the expected number of random mutations per individual per generation. Evolutionary dynamics implicitly assume $U \ll 1$, so that offspring remain close to parental genotypes in configuration space.

G.3 The Error Threshold (Information-Theoretic Formulation)

Let:

I_v = minimum information required to specify a viable configuration under boundary conditions B (equivalent to $F_{\min}$) I_m = information destroyed by mutation per generation

For a genome of length L over an alphabet of size k :

$$I_m \approx U \cdot \log_2(k)$$

Darwinian evolution requires:

$$I_m < I_v$$

If:

$$I_m \geq I_v$$

then mutation erases more information per generation than selection can preserve. Viability cannot be maintained, regardless of population size or selection strength.

This condition defines a hard operational ceiling on random variation.

G.4 Selection Cannot Compensate Above the Threshold

Selection acts only on viable configurations. When U exceeds the threshold:

- offspring are overwhelmingly non-viable
- fitness gradients collapse
- heredity loses coherence

Formally, let $V(B) \subset X$ be the viability set under boundary B . For $U > U_c$:

$$P(x_{t+1} \in V(B) \mid x_t \in V(B)) \rightarrow 0$$

Selection has no domain of action once heredity is statistically destroyed. This is not slow degeneration; it is categorical failure.

G.5 Phase Interpretation

The mutation threshold represents a phase boundary between two regimes:

Regime I (Darwinian): • $U < U_c$ • heredity preserved • selection meaningful • gradual adaptation possible

Regime II (Post-Darwinian): • $U \geq U_c$ • heredity decoheres • selection undefined • only structural phase transitions possible

Crossing this threshold forces the system out of evolutionary dynamics entirely.

G.6 Relation to Boundary Discontinuities

Boundary shifts typically increase effective mutation load:

- radiation increases μ
- chemical stress increases error rates
- thermal stress destabilizes replication

Thus, extreme environments induce a double failure:

1. boundary displacement removes prior viability sets
2. mutation flux exceeds the Darwinian operating range

Even if a continuous path existed in principle, it would be erased faster than it could be traversed.

G.7 Why Waiting Does Not Help

Time does not rescue systems above the mutation threshold. Let T be the available evolutionary time.

If $U \geq U_c$:

Expected viable lineage length $\approx O(1)$

No accumulation is possible. Evolution does not slow down — it stops being defined.

G.8 Integration with TNA and $F_{\min}$

Within the TNA framework:

- $F_{\min}$ defines the minimal coherence required for viability
- mutation supplies random exploration

Darwinian evolution operates only when:

$\text{mutation_flux} < F_{\min}$

Beyond this point, randomness destroys coherence faster than selection can filter it. Survival then depends on pre-aligned structure or discrete reconfiguration admitted by new boundary conditions.

G.9 Formal Statement — The Mutation Threshold Theorem

Theorem (Darwinian Operating Threshold):

For any evolutionary system with minimum viable information $F_{\min}$, there exists a critical mutation rate U_c such that if the mutation flux exceeds U_c , Darwinian evolution cannot preserve viability, independent of selection strength, population size, or time.

Above this threshold, evolutionary dynamics collapse into either extinction or non-Darwinian phase transition.

G.10 Structural Conclusion

Evolution is doubly local:

- local in boundary space (fails at discontinuities)
- local in mutation space (fails above U_c)

Beyond either limit, Darwinian mechanisms are not merely insufficient — they are undefined. What remains is not accelerated evolution, but structural phase change governed by coherence constraints rather than stochastic search.

Darwinian evolution is not merely limited by environmental continuity, but by an intrinsic mutation-rate threshold. Beyond this threshold, heredity decoheres faster than selection can act, rendering evolutionary dynamics undefined. What follows is not rapid evolution, but structural phase transition.

Appendix — Quantitative Limits of Darwinian Probability in the Cambrian Explosion

Explaining the Cambrian Explosion by fossil continuity alone is like placing a row of candles and expecting them to generate lightning. No amount of incremental illumination can reproduce a discharge that belongs to a different physical regime.

The Cambrian Explosion as a Probability Collapse Event

The term *Cambrian Explosion* is often treated as an unfortunate metaphor. From the perspective of the **Theorem of Axiomatic Necessity (TNA)**, it is instead a literal description of a **systemic phase transition** in which stochastic search (N_0) becomes mathematically irrelevant.

1. Configuration Space and the Thermal Death of Random Search

To account for the Cambrian Explosion via random mutation plus selection, the following must hold:

1.1 Configuration Space Size

Let:

- L : effective genomic length relevant for a new body plan (order 10^5 – 10^6 base pairs),
- alphabet size $k = 4$.

The raw configuration space is:

$$|\Omega| = k^L \approx 4^{10^6} \approx 10^{600,000}$$

Even if we restrict attention to **functional subsets** (protein-coding, regulatory coherence), conservative estimates still give:

$$|\Omega_{\text{functional}}| \gtrsim 10^{10^3}$$

This is the search space for **one** new phylum-level architecture.

2. Time and Information Throughput Constraints

2.1 Available Time

Cambrian diversification occurred over approximately:

$$T \approx 20 \text{ million years} \approx 6.3 \times 10^{14} \text{ seconds}$$

2.2 Maximum Mutation Trials

Let:

- population size $N \sim 10^{10}$ (optimistic),
- mutation rate per genome per generation $\mu_g \sim 10^{-3}$,
- generation time $\tau \sim 1$ year.

Total trials:

$$\text{Trials} \approx N \cdot \mu_g \cdot \frac{T}{\tau} \approx 10^{10} \cdot 10^{-3} \cdot 2 \times 10^7 \approx 2 \times 10^{14}$$

This is **14 orders of magnitude**.

The search space is **thousands of orders of magnitude larger**.

The system is information-starved by construction.

3. Simultaneity Problem: Multiple Body Plans

The Cambrian Explosion produced **~30 distinct phyla**, not sequentially but **overlapping in time**.

If P_1 is the probability of discovering one viable body plan:

$$P_{\text{total}} = P_1^{30}$$

Given $P_1 \ll 10^{-100}$, the joint probability is effectively zero:

$$P_{\text{total}} \rightarrow 0 \quad \text{faster than exponential}$$

This is not improbability — it is **combinatorial impossibility**.

4. Boundary Stress Paradox

Darwinian stochastic optimization requires:

- relatively stable boundary conditions,
- viable intermediates,
- time for gradient ascent.

But the Cambrian period coincides with **violent boundary shifts**:

- oxygenation events,
- ocean chemistry restructuring,
- tectonic and ecological instability.

Under such stress:

- mutation lethality increases,
- selectable intermediates vanish,
- fitness gradients flatten or fragment.

Formally:

$$\nabla F \rightarrow 0 \quad \text{or undefined}$$

Yet empirically, **complexity increases**.

This is the opposite of what stochastic search predicts.

5. TNA Interpretation: Syntony, Not Search

Under TNA, the Cambrian Explosion is not a random walk but a **coherence transition**.

Let:

- N_0 : stochastic variation,
- N_1 : external coherence regime defined by boundary conditions,

- $F_{\min}$: minimal viable coherence under new constraints.

When boundary conditions shift discontinuously:

- previous configurations become non-viable,
- random exploration collapses,
- only configurations aligned with the new $F_{\min}$ persist.

This is not “trial and error” but **forced crystallization**.

Mathematically, the system undergoes:

$$\Omega(B_t) \longrightarrow \Omega(B_{t+1}), \quad \Omega(B_t) \cap \Omega(B_{t+1}) \approx \emptyset$$

The outcome is a **quantized jump** in organization.

6. The Eye Problem (Explicit Probability Bound)

Probability of assembling a functional eye via random mutation in 10^7 years:

$$P_{\text{eye}} \approx 0$$

Not small — **zero within physical limits**.

But if visual predation becomes a **structural necessity** under new boundary conditions:

$$P_{\text{eye}} = 1 \quad (\text{axiomatic})$$

Eyes appear because **they must**, not because they were found.

7. Why the Calculation Is Avoided

Standard evolutionary theory implicitly assumes:

- continuity,
- dense intermediates,
- computable gradients.

The Cambrian Explosion violates all three.

Performing the probability calculation exposes a contradiction:
the required information throughput exceeds the biological substrate's capacity.

Thus, the calculation is not controversial — **it is lethal**.

8. Conclusion: Chance Is the Noise of Calm Regimes

Random mutation optimizes only when:

- pressure is low,
- constraints are stable,
- time is abundant.

Under extreme boundary shifts, chance collapses.

The Cambrian Explosion demonstrates that:

- life does not *search* new architectures,
- life *locks into* new coherence regimes imposed by necessity.

What appears as “explosion” is the **visible footprint of a forced phase transition**.

*The fossil record shows immediate functionality, not gradual failure.
Chance does not produce that.
Constraint-driven system reconfiguration does.*

Corollary (The Cambrian Phase Transition Corollary)

Statement.

Let N_0 be a biological system operating under Darwinian dynamics (random variation + local selection) within a fixed regime of boundary conditions B . Let $\Omega(B)$ denote the space of viable biological configurations under B , and let $F_{\min}(B)$ be the minimal coherence required for persistence.

If the boundary conditions undergo a discontinuous shift $B \rightarrow B'$ such that:

1. The dimensionality or structural complexity of $\Omega(B')$ increases discontinuously, and
2. The rate of random variation required to explore $\Omega(B')$ exceeds the information-processing capacity of the biological substrate,

then Darwinian evolution in N_0 becomes non-operative as a generative mechanism.

In this regime, the emergence of new biological architectures cannot be described as a gradual accumulation of intermediate forms. Instead, system survival requires a **phase transition**, in which viable configurations crystallize directly at or above $F_{\min}(B')$.

Implication.

The Cambrian Explosion is not an extreme case of accelerated Darwinian evolution, but a transition between coherence regimes. The absence of dense transitional fossil sequences is not an artifact of preservation or sampling, but a necessary consequence of a discontinuous shift in the system's boundary conditions.

Interpretative Remark.

Attempting to explain the Cambrian Explosion solely through fossil continuity is therefore a category error: it treats a phase transition as if it were an incremental process.

Explaining the Cambrian Explosion by arranging fossil intermediates is like lining up candles and expecting them to produce lightning. No accumulation of small signals can reproduce a discharge that belongs to a different physical regime.